

Lab: The Teenage Brain -- Synaptic Plasticity and How You Learn

Objective

Investigate the biological mechanisms of **learning** -- synaptic plasticity, memory formation, and the unique properties of the adolescent brain -- through the lens of Active Inference model updating.

Prerequisites

- Completed Biology modules on Systems through Action
- Read the module on Synaptic Plasticity, Memory, and the Teenage Brain

Part 1: Synaptic Plasticity Mechanism

Goal: Understand how synapses change to encode new learning.

Research Long-Term Potentiation (LTP) and fill in the mechanism:

Step	What Happens at the Synapse	Active Inference Interpretation
1. Repeated activation		Repeated prediction errors at this connection
2. NMDA receptor activation		Error signal reaches the synapse
3. Calcium influx		Triggers model update
4. AMPA receptor insertion		Strengthened connection = updated parameter
5. Structural change		Permanent model revision

What is the opposite of LTP? (Long-Term Depression -- LTD.) When would the brain need to weaken a connection?

Write your response here

Part 2: Memory Systems Comparison

Goal: Map different memory types to different model components.

Memory Type	Brain Region	Duration	Example	What Part of the Model?
Working memory				
Episodic memory				
Procedural memory				
Semantic memory				

How does sleep help consolidate memories? (Hint: the hippocampus "replays" experiences to the cortex during sleep, transferring learning.)

Write your response here

Part 3: Your Teenage Brain -- A Special Window

Goal: Understand why adolescence is a critical period for learning.

Research the teenage brain and answer:

1. **Synaptic pruning:** Your brain is actively removing unused connections. Why is this beneficial for the generative model? (Hint: it increases efficiency by keeping only the connections that have been used.)
2. **Prefrontal cortex development:** This region is the last to mature (not until mid-20s). How does this affect planning and impulse control?
3. **Heightened plasticity:** Teenage brains are more plastic than adult brains. What are the advantages and risks of high plasticity?
4. **Sleep and learning:** Why do teenagers need more sleep than adults for optimal learning?

Write your response here

Part 4: Learning and Health

Goal: Connect learning biology to health behaviors.

How do the following affect synaptic plasticity and model updating?

Factor	Effect on Learning	Biological Mechanism
Regular exercise		
Chronic stress		
Sleep deprivation		
Nutrition		
Social connection		

Write your response here

Summary Table

Concept	Biological Mechanism	Active Inference Connection
Synaptic plasticity	LTP/LTD at synapses	Model parameter updating
Memory consolidation	Hippocampal replay during sleep	Transferring learning to long-term model
Synaptic pruning	Removal of unused connections	Model simplification and efficiency
Critical periods	Windows of heightened plasticity	Times when the model updates most rapidly